Fiscal Year:	FY 2004	**Task Last Updated:**	FY 06/15/2004
PI Name:	Leff, Laura Ph.D.		
Project Title:	The Effects of Microgravity on the Ecology of Biofilm Bacteria		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Evolutionary biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Microbiology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	44242	**Congressional District:**	17
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2000 Space Biology 00-OBPR-01
Start Date:	07/01/2001	**End Date:**	06/30/2005
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	1
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA ARC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NAG 2-1497		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	Throughout this study several types of methods will be employed to assess bacterial community composition and gauge the responses of this community. Several molecular methods are available for our use: sequencing of DNA, PCR and subsequent analysis by denaturing gradient gel electrophoresis, and fluorescent in situ hybridization. Each allows us to characterize bacterial communities using culture-independent methods. In addition to these molecular methods, we also will rely on direct enumeration using epifluorescent microscopy and cultivation. Methods will be applied in varying combinations to material from three basic sources: 1) bacteria from the JSC culture collection, 2) bacterial samples collected from space shuttle/station water systems, and 3) samples collected during our ground based experimentation. By applying methods in combination, we can understand the ecology of microbes in this system. To examine responses of bacteria to simulated microgravity and study biofilm formation by bacteria from space station water systems. Formation of biofilms in water systems may enhance bacterial survival and increase the potential for negative impacts		

	on water quality.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Bacteria are chronic inhabitants of water systems, including those associated with NASA activities. There are two primary objectives for this study. First, we are using state of the art molecular methods to assess bacterial community composition in water systems. This includes use of 16S rDNA sequencing and subsequent phylogenetic analysis to identify and characterize bacteria isolated from various systems (including space shuttle water systems, bacteria previously isolated from Mir, etc.). We will use our molecular ecology expertise to examine bacteria that defy cultivation (bacteria under oligotrophic conditions, like in pure water, are typically difficult to culture and thus other methods must be employed to examine this important group). Second, using bacteria isolated previously from the JSC collection and any new isolates, we are conducting ground based experiments on factors that affect biofilm formation and particularly the influence of microgravity on the growth and success of bacteria. This will be accomplished using slow turning lateral vessels to simulate microgravity. The overall significance is that this will provide the basic information needed to understand factors that influence the number and type of bacteria found in shuttle and space station water systems.
Task Progress:	Task progress for current year: Bacteria originally isolated from water tanks aboard the Mir and International space stations were provided by JSC. In the previous report, studies on the Mir isolates using modeled microgravity when the cells were resuspended in water, 0.002% (w/v) nutrient broth and 0.2% (w/v) nutrient broth were described. These results were obtained using specific operational conditions to simulate microgravity using a clinostat, and Escherichia coli was used to investigate parameters such as rotation speed, the effect of sampling and the type of medium used. E. coli was chosen because many studies on this species have been performed under actual microgravity and modeled microgravity conditions. The results of these studies showed that significant differences were obtained at rotation speeds between 30-50 rpm only. Removing samples every 24 h had no effect on the results obtained and thus the act of sample collection was determined not to significantly alter the results. When E. coli was grown in nutrient broth the abundance of cells was slightly higher and the cells were smaller under modeled microgravity compared to the normal gravity controls. These observations were not apparent when E. coli was grown in minimal medium. These results may be useful to researchers working with microgravity in modeled microgravity conditions. The aim of the next study was to investigate the attachment of bacterial cells growing in a model microgravity environment to stainless steel disks. The Mir space station isolates were grown in either 0.002% (w/v) or 0.2% (w/v) nutrient broth. Planktonic cells were enumerated using DAPI and bacteria attaching to the steel disks were removed using the stomacher and enumerated using the Live/Dead BacLight kit and DAPI. The majority of results showed no significant differences in attachment under simulated microgravity compared to normal gravity. However, a Pseudomonas fluorescens isolate grown in 0.2% (w/v) nutrient broth showed significantly higher attachment under microgravity compared to normal gravity. This may be attributed to some unknown change in the cell physiology because planktonic cells showed no significant differences under the two different gravity conditions. The other bacterial isolate studied in the same experiment showed significantly higher planktonic cell counts under simulated microgravity compared to normal gravity, consequently resulting in higher attachment under modeled microgravity. These results confirm that biofilms can be formed under reduced gravity conditions. In another study, the attachment, detachment and long-term persistence in biofilms of the Mir and ISS isolates. Stainless steel disks were used as the substrate for biofilm formation. The bacteria were enumerated using Live/Dead BacLight kit and DAPI. Bacteria grown in nutrient broth attached at a higher abundance than those grown in minimal medium because the number of the cells present in nutrient broth was higher. Detachment of bacteria showed no significant differences between types of isolates studied and type of medium used. However, it was evident that a significant proportion of the bacterial cells had detached from the steel surface. Studies over a ten-week period showed that the number of detached and attached cells remained consistent and without loss of viability. These results provide an insight into how bacteria isolated from water tanks aboard the space stations can continue to survive for long periods of time. A different study was performed to investigate the effect of magnesium ions on the formation of biofilms. A P. fluorescens isolate from the Mir space station was used in this study. The biofilms were created in minimal medium under static conditions. The number of cells attaching to glass coverslips was enumerated using DAPI, hydrophobicity was determined and the biofilms were viewed directly using confocal microscope in combination with Live/Dead BacLight staining. The results showed that higher magnesium ion concentrations lead to an increase in the attachment of bacteria to the glass coverslips. These results indicate that the corrosion products of stainless steel may accelerate the formation of biofilms.
Bibliography Type:	Description: (Last Updated: 04/09/2020)
Articles in Peer-reviewed Journals	Baker PW, Leff L. "The effect of simulated microgravity on bacteria from the Mir Space Station." Microgravity Sci Technol. 2004 Mar;15(1):35-41. https://doi.org/10.1007/BF02870950 ; PubMed PMID: 15773020 , Mar-2004
Dissertations and Theses	Song B. "Identification and characterization of bacterial isolates from the Mir space station and the impact of Mg2+ concentration on biofilm formation of Pseudomonas fluorescens." MS Thesis, Kent State University, May 2004. , May-2004
Presentation	Baker, P.W.; Meyer, M.L.; Leff, L.G. "Responses of Escherchia coli ATCC 26 to simulated microgravity." American Society for Microbiology, May 2004, New Orleans LA. May-2004
Presentation	Meyer, M.L.; Kuny, V.; Baker, P.W.; Leff, L.G. "Biofilm formation by bacteria from the Mir Space Station to stainless steel disks." American Society for Microbiology, May 2004, New Orleans LA. May-2004
Presentation	Song, B.; Leff, L.G. "Effects of Ca+2 and Mg+2 on biofilm formation of Pseudomonas fluorescens in static conditions." American Society for Microbiology, May 2004, New Orleans LA. May-2004
Presentation	Song, B.; Leff, L.G. "Identification and characterization of bacterial strains from the Mir Space Station." Ohio Branch Meeting American Society for Microbiology, 2004, Youngstown, OH. Apr-2004

Presentation	Baker, P.W.; Leff, L.G. "Bacterial attachment to stainless steel disks in a simulated microgravity environment." Ohio Branch Meeting American Society for Microbiology, 2004, Youngstown, OH. Apr-2004
Presentation	Baker, P.W.; Leff, L.G. "Bacterial growth under simulated microgravity conditions." American Society for Gravitational and Space Biology Annual Meeting. Nov-2003